



HelioSolv

POWERING A CIRCULAR FUTURE



REUSE



RECOVER



REGENERATE



Team Fluxara

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Solar Energy Has a Waste Problem

Clean energy today. A growing waste challenge tomorrow.



**~600
kilotonnes**

of solar PV waste
expected by 2030
in India.



**~11,221
kilotonnes**

projected by 2047.



**Valuable
materials**

like silicon, silver
and glass are often
lost without
proper recovery.



Many end-of-life panels
end up in landfills,
leading to loss of valuable
resources and environmental risks.



Solar panels deliver
clean energy for **25–30 years**
and power a greener tomorrow.

Waste Today. Resource Tomorrow.

End-of-life solar panels aren't just waste — they're a source of valuable resources.



Reuse

Extend panel life
where possible



Recover

Extract valuable
materials



Create Value

Local jobs &
circular businesses

1 SOLAR PANELS IN USE

4 NEW OPPORTUNITIES

2 REUSE WHEN POSSIBLE

3 RECOVER VALUABLE MATERIALS

A
CIRCULAR
FUTURE
FOR SOLAR

REDUCE WASTE

Divert panels from
landfills

RECOVER RESOURCES

Value from silver,
silicon, glass and more

ENABLE OPPORTUNITIES

Local MSME-led
recycling ecosystems



Silver



Silicon



Glass



Others

Introducing HelioSolv

An AI-assisted, reuse-first platform for smarter solar panel recovery.

1

AI-Assisted Assessment

Identify panel condition using computer vision



2

Reuse-First Decisions

Evaluate for reuse, further testing or recycling



3

Material Recovery

Recover valuable materials using green chemistry (DES concept)



4

Decentralized Micro-Plants

Enable local MSMEs with scalable, modular recycling facilities

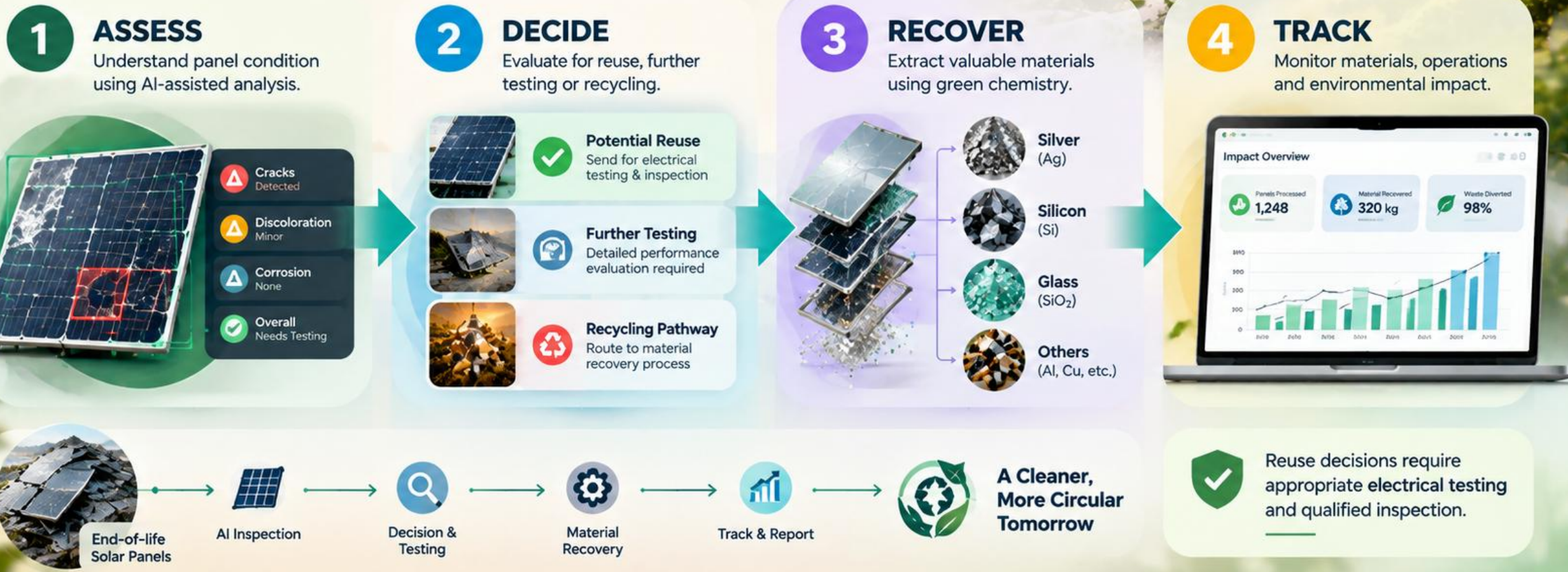


From Assessment to Recovery

A simple, integrated workflow for a circular solar future.



*Assess.
Decide.
Recover.
Track.*

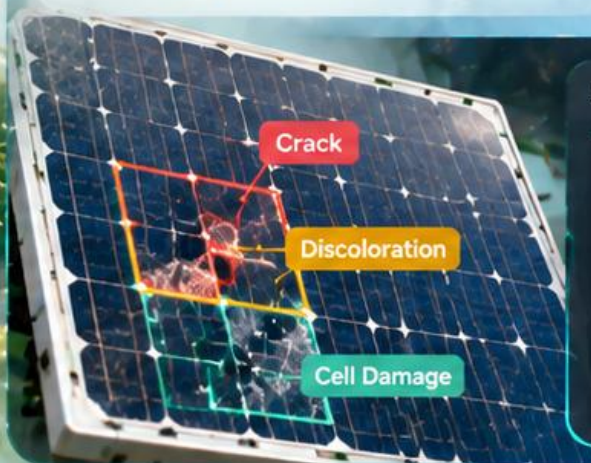


Smarter Decisions. Circular Technology.

Combining AI and green chemistry to unlock a circular future for solar panels.

1 Computer Vision

AI-assisted visual screening of panel condition



AI Analysis Result

78%
Confidence

-  **Cracks**
Detected
-  **Discoloration**
Detected
-  **Cell Damage**
Minor
-  **Overall Condition**
Needs Inspection

2

Proposed DES-Based Silver Recovery

Using Deep Eutectic Solvents (Choline Chloride + Ethylene Glycol)



Silver (Ag) Recovery



~80°C
Mild operating conditions



Lower environmental impact (vs. conventional methods)



Research-stage process (Experimental validation required)



AI-Assisted Assessment

Identify visible damage and suggest next steps.



Reuse-First Decision Support

Evaluate for reuse, further testing or recycling.



Green Chemistry Innovation

Proposed DES-based silver recovery.



Digital Traceability

Track materials, batches and impact across the value chain.

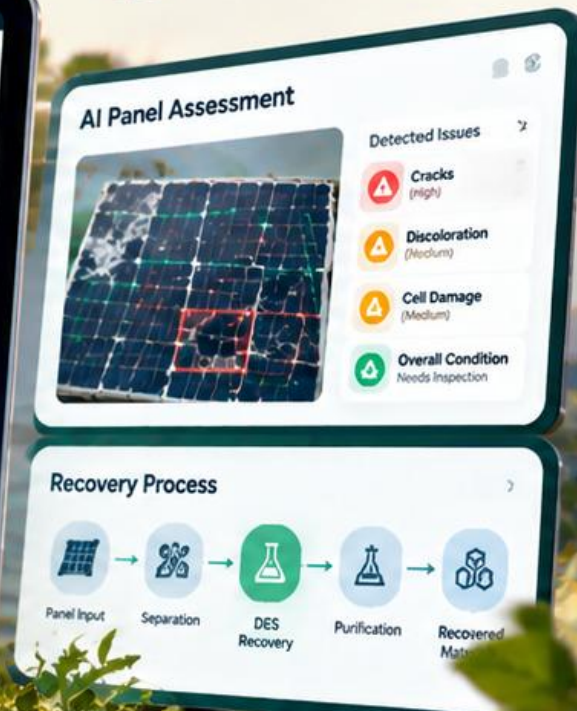
Experience HelioSolv

An interactive platform for a circular solar future.



Scan to Explore
Live Prototype

<https://heliosolv.vercel.app>



1

AI-Powered Assessment

Upload panel images & get instant analysis



2

Reuse-First Decision Support

Evaluate for reuse, testing or recycling



3

Recovery & Inventory

Track material recovery and manage batches



4

Impact Analytics

Monitor environmental and operational impact



Local Plants. Circular Value.

A scalable, MSME-led model to turn solar waste into economic and environmental value.



Decentralized Micro-Plants

Closer to waste sources



MSME-Led Operations

Local jobs & entrepreneurship



Material Recovery

Silver, silicon, glass and more



Replicable Model

Scale across regions

1 Collection Points

From solar farms, installers & industries

2 Local Aggregation

Sorting & initial screening

3 Micro-Plant Processing

Reuse, recovery & material extraction

4 Reuse Market

Refurbished panels for second life

5 Recovered Materials

Supply to industries (e.g. glass, silicon, silver)

Scalable Expansion Model



Pilot Plant
(Year 1)
1 region



Regional Plants
(Years 2-3)
5+ regions



National Scale
(Years 4-5)
20+ regions



Value for Stakeholders



Cleaner Environment



Local Jobs & MSME Growth



Resource Value Recovery



Replicable & Investable Model

Revenue Streams (Est.)



Sale of Refurbished Panels
Second-life panels to local markets

₹₹₹



Recovered Materials
Silver, silicon, glass, aluminium, etc.

₹₹₹



Recycling Services
Panel collection & processing fees

₹₹



ESG & Compliance Support
EPR and sustainability reporting

₹

Note: All financial figures are estimates and for illustrative purposes only. Actuals may vary based on location, scale and market conditions.

From Local Action to a National Circular Network.

Designed for Measurable Impact.

Turning solar waste into a cleaner, more circular tomorrow.



Lower Emissions

~1.2 t CO₂e

avoided per ton of panels recovered



Higher Circularity

70–90%

material recovery potential (est.)



Less Landfill Waste

Up to 80%

diversion through reuse-first approach

1

VALIDATE
(0–1 Year)



Test and refine the technology.



Experimental validation of DES-based recovery



Improve AI assessment accuracy



Refine operational model

2

PILOT
(1–3 Years)



Set up and operate pilot micro-plants.



1–3 pilot micro-plants



MSME-led operations



Validate recovery efficiency and unit economics



Build supply chain network

3

SCALE
(3–5+ Years)



Expand to multiple regions.



20+ micro-plants across India (est.)



Thousand+ local jobs



Large-scale material recovery & waste diversion



Replicable model for global markets



A cleaner environment. Stronger communities. A circular solar future.



HelioSolv
POWERING A CIRCULAR FUTURE

Thank You

Let's Give Every Solar Panel
a **Better Tomorrow.**



Try Our **Prototype**



<https://heliosolv.vercel.app>



Team **Fluxara**



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